

Unit 40: Integration by Partial Fractions — Full Solutions

Warm-up

1. $\int \frac{dx}{(x-1)(x+1)}$

Decompose: $\frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$. Clear denominators: $1 = A(x+1) + B(x-1)$.

$x = 1$: $1 = 2A \Rightarrow A = \frac{1}{2}$. $x = -1$: $1 = -2B \Rightarrow B = -\frac{1}{2}$.

$$\int \frac{dx}{(x-1)(x+1)} = \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$$

2. $\int \frac{dx}{(x-2)(x+3)}$

$1 = A(x+3) + B(x-2)$. $x = 2$: $1 = 5A \Rightarrow A = \frac{1}{5}$. $x = -3$: $1 = -5B \Rightarrow B = -\frac{1}{5}$.

$$= \frac{1}{5} \ln|x-2| - \frac{1}{5} \ln|x+3| + C$$

3. $\int \frac{3x+5}{(x+1)(x+2)} dx$

$3x+5 = A(x+2) + B(x+1)$. $x = -1$: $2 = A \Rightarrow A = 2$. $x = -2$: $-1 = -B \Rightarrow B = 1$.

$$= 2 \ln|x+1| + \ln|x+2| + C$$

4. $\int \frac{5x-1}{(x-1)(x-3)} dx$

$5x-1 = A(x-3) + B(x-1)$. $x = 1$: $4 = -2A \Rightarrow A = -2$. $x = 3$: $14 = 2B \Rightarrow B = 7$.

$$= -2 \ln|x-1| + 7 \ln|x-3| + C$$

5. $\int \frac{dx}{x^2-4}$

$x^2-4 = (x-2)(x+2)$. $1 = A(x+2) + B(x-2)$. $x = 2$: $A = \frac{1}{4}$. $x = -2$: $B = -\frac{1}{4}$.

$$= \frac{1}{4} \ln|x-2| - \frac{1}{4} \ln|x+2| + C$$

6. $\int \frac{x+7}{x^2+x-6} dx$

Factor: $x^2+x-6 = (x+3)(x-2)$. $x+7 = A(x-2) + B(x+3)$. $x = -3$: $4 = -5A \Rightarrow A = -\frac{4}{5}$. $x = 2$: $9 = 5B \Rightarrow B = \frac{9}{5}$.

$$= -\frac{4}{5} \ln|x+3| + \frac{9}{5} \ln|x-2| + C$$

Standard

7. $\int \frac{dx}{(x-1)^2(x+1)}$

Decompose: $\frac{1}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$.

Clear: $1 = A(x-1)(x+1) + B(x+1) + C(x-1)^2$.

$x = 1$: $1 = 2B \Rightarrow B = \frac{1}{2}$. $x = -1$: $1 = 4C \Rightarrow C = \frac{1}{4}$.

Plug $x = 0$ to find A : $1 = A(-1)(1) + B(1) + C(1) = -A + B + C = -A + \frac{1}{2} + \frac{1}{4} = -A + \frac{3}{4}$. So $A = \frac{3}{4} - 1 = -\frac{1}{4}$.

$$\int \frac{dx}{(x-1)^2(x+1)} = -\frac{1}{4} \ln|x-1| - \frac{1}{2(x-1)} + \frac{1}{4} \ln|x+1| + C$$

8. $\int \frac{3x-1}{x^2(x+1)} dx$

Decompose: $\frac{3x-1}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}$.

Clear: $3x-1 = Ax(x+1) + B(x+1) + Cx^2$.

$x = 0$: $-1 = B \Rightarrow B = -1$. $x = -1$: $-4 = C \Rightarrow C = -4$.

Plug $x = 1$: $2 = A(1)(2) + B(2) + C(1) = 2A + 2B + C = 2A - 2 - 4 = 2A - 6$. So $2A = 8 \Rightarrow A = 4$.

$$\int \frac{3x-1}{x^2(x+1)} dx = 4 \ln|x| + \frac{1}{x} - 4 \ln|x+1| + C$$

9. $\int \frac{dx}{(x-1)(x-2)(x-3)}$

Decompose: $\frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$. Clear: $1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$.

$x = 1$: $1 = 2A \Rightarrow A = \frac{1}{2}$. $x = 2$: $1 = -B \Rightarrow B = -1$. $x = 3$: $1 = 2C \Rightarrow C = \frac{1}{2}$.

$$= \frac{1}{2} \ln|x-1| - \ln|x-2| + \frac{1}{2} \ln|x-3| + C$$

10. $\int \frac{2x+1}{(x-2)^2(x+1)} dx$

Decompose: $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+1}$. Clear: $2x+1 = A(x-2)(x+1) + B(x+1) + C(x-2)^2$.

$x = 2$: $5 = 3B \Rightarrow B = \frac{5}{3}$. $x = -1$: $-1 = 9C \Rightarrow C = -\frac{1}{9}$.

Plug $x = 0$: $1 = A(-2)(1) + B(1) + C(4) = -2A + B + 4C = -2A + \frac{5}{3} - \frac{4}{9}$.

Convert to ninths: $\frac{5}{3} = \frac{15}{9}$, so $-2A + \frac{15}{9} - \frac{4}{9} = -2A + \frac{11}{9}$. Set equal to $1 = \frac{9}{9}$: $-2A = \frac{9}{9} - \frac{11}{9} = -\frac{2}{9} \Rightarrow A = \frac{1}{9}$.

$$\int \frac{2x+1}{(x-2)^2(x+1)} dx = \frac{1}{9} \ln|x-2| - \frac{5}{3(x-2)} - \frac{1}{9} \ln|x+1| + C$$

11. Long division: $\frac{x^3 + 2x^2 - 5}{x+3}$

$$x^3 + 2x^2 + 0x - 5 \div (x+3)$$

$x^3 \div x = x^2$; $x^2(x+3) = x^3 + 3x^2$; subtract: $(x^3 + 2x^2) - (x^3 + 3x^2) = -x^2$, bring down $0x$: $-x^2 + 0x$.

$-x^2 \div x = -x$; $-x(x+3) = -x^2 - 3x$; subtract: $(-x^2 + 0x) - (-x^2 - 3x) = 3x$, bring down -5 : $3x - 5$.

$3x \div x = 3$; $3(x+3) = 3x + 9$; subtract: $(3x - 5) - (3x + 9) = -14$.

$$\frac{x^3 + 2x^2 - 5}{x+3} = x^2 - x + 3 - \frac{14}{x+3}$$

12. $\int \frac{x^2+1}{x-1} dx$

Long division: $x^2 + 0x + 1 \div (x-1)$.

$x^2 \div x = x$; $x(x-1) = x^2 - x$; subtract: $(x^2 + 0x) - (x^2 - x) = x$, bring down 1 : $x + 1$.

$x \div x = 1$; $1(x-1) = x - 1$; subtract: $(x + 1) - (x - 1) = 2$.

$$\frac{x^2+1}{x-1} = x + 1 + \frac{2}{x-1}$$

$$\int \frac{x^2+1}{x-1} dx = \int \left(x + 1 + \frac{2}{x-1} \right) dx = \frac{x^2}{2} + x + 2 \ln|x-1| + C$$

Challenge

13. $\int \frac{2x-6}{x^2-6x+8} dx$

Notice: $\frac{d}{dx}[x^2 - 6x + 8] = 2x - 6$ — an exact match with the numerator! No partial fractions needed:

$$\int \frac{2x-6}{x^2-6x+8} dx = \ln|x^2 - 6x + 8| + C$$

(Since $x^2 - 6x + 8 = (x-2)(x-4)$, this is equivalent to $\ln|(x-2)(x-4)| + C$.)

14. $\int \frac{2x-7}{x^2-7x+12} dx$

Notice: $\frac{d}{dx}[x^2 - 7x + 12] = 2x - 7$ — another exact match!

$$\int \frac{2x-7}{x^2-7x+12} dx = \ln|x^2-7x+12| + C$$

(Equivalent to $\ln|(x-3)(x-4)| + C$.)

$$15. \int \frac{2x+6}{(x+1)(x-1)^2} dx$$

This numerator does not match the derivative of the denominator, so genuine partial fractions are needed.

Decompose: $\frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$. Clear: $2x+6 = A(x-1)^2 + B(x+1)(x-1) + C(x+1)$.

$x = -1$: $4 = 4A \Rightarrow A = 1$. $x = 1$: $8 = 2C \Rightarrow C = 4$.

Plug $x = 0$: $6 = A(1) + B(-1) + C(1) = A - B + C = 1 - B + 4 = 5 - B$. So $B = 5 - 6 = -1$.

$$\int \frac{2x+6}{(x+1)(x-1)^2} dx = \ln|x+1| - \ln|x-1| - \frac{4}{x-1} + C$$

$$16. \int \frac{x+1}{(x-1)(x-2)(x-3)} dx$$

Decompose: $\frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$. Clear: $x+1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$.

$x = 1$: $2 = 2A \Rightarrow A = 1$. $x = 2$: $3 = -B \Rightarrow B = -3$. $x = 3$: $4 = 2C \Rightarrow C = 2$.

$$\int \frac{x+1}{(x-1)(x-2)(x-3)} dx = \ln|x-1| - 3\ln|x-2| + 2\ln|x-3| + C$$

$$17. \int \frac{x^3 + 3x^2 + 2x - 7}{x+2} dx$$

Long division: $x^3 + 3x^2 + 2x - 7 \div (x+2)$.

$x^3 \div x = x^2$; $x^2(x+2) = x^3 + 2x^2$; subtract: $(x^3 + 3x^2) - (x^3 + 2x^2) = x^2$, bring down $2x$: $x^2 + 2x$.

$x^2 \div x = x$; $x(x+2) = x^2 + 2x$; subtract: $(x^2 + 2x) - (x^2 + 2x) = 0$, bring down -7 : $0 - 7 = -7$.

$$\frac{x^3 + 3x^2 + 2x - 7}{x+2} = x^2 + x - \frac{7}{x+2}$$

$$\int \frac{x^3 + 3x^2 + 2x - 7}{x+2} dx = \frac{x^3}{3} + \frac{x^2}{2} - 7\ln|x+2| + C$$

Applied

$$18. \int \frac{dx}{(x-2)(x-5)}$$

$$1 = A(x-5) + B(x-2). \quad x = 2: A = -\frac{1}{3}. \quad x = 5: B = \frac{1}{3}.$$

$$= -\frac{1}{3} \ln|x-2| + \frac{1}{3} \ln|x-5| + C = \frac{1}{3} \ln \left| \frac{x-5}{x-2} \right| + C$$

$$19. \int \frac{dP}{P(100-P)}$$

$$\text{Decompose: } \frac{A}{P} + \frac{B}{100-P}. \quad 1 = A(100-P) + BP. \quad P = 0: A = \frac{1}{100}. \quad P = 100: B = \frac{1}{100}.$$

$$= \frac{1}{100} \ln|P| - \frac{1}{100} \ln|100-P| + C = \frac{1}{100} \ln \left| \frac{P}{100-P} \right| + C$$

(This is exactly the building-block integral needed to solve the logistic growth differential equation, $\frac{dP}{dt} = kP(100-P)$.)

$$20. \int_0^2 \frac{x+3}{(x+1)(x+2)} dx$$

$$\text{Decompose: } \frac{A}{x+1} + \frac{B}{x+2}. \quad x+3 = A(x+2) + B(x+1). \quad x = -1: A = 2. \quad x = -2: B = -1.$$

$$\text{Antiderivative: } 2 \ln|x+1| - \ln|x+2|.$$

$$\text{At } x = 2: 2 \ln 3 - \ln 4. \quad \text{At } x = 0: 2 \ln 1 - \ln 2 = 0 - \ln 2 = -\ln 2.$$

$$(2 \ln 3 - \ln 4) - (-\ln 2) = 2 \ln 3 - \ln 4 + \ln 2$$

$$\text{Simplify using } \ln 4 = 2 \ln 2: 2 \ln 3 - 2 \ln 2 + \ln 2 = 2 \ln 3 - \ln 2 = \ln 9 - \ln 2 = \ln \left(\frac{9}{2} \right).$$

$$\text{Answer: } \ln \left(\frac{9}{2} \right) \approx 1.504.$$

$$21. \int_4^5 \frac{dx}{(x-3)(x-6)}$$

$$1 = A(x-6) + B(x-3). \quad x = 3: A = -\frac{1}{3}. \quad x = 6: B = \frac{1}{3}.$$

$$\text{Antiderivative: } -\frac{1}{3} \ln|x-3| + \frac{1}{3} \ln|x-6| = \frac{1}{3} \ln \left| \frac{x-6}{x-3} \right|.$$

$$\text{At } x = 5: \frac{1}{3} \ln \left| \frac{-1}{2} \right| = \frac{1}{3} \ln \left(\frac{1}{2} \right) = -\frac{\ln 2}{3}.$$

$$\text{At } x = 4: \frac{1}{3} \ln \left| \frac{-2}{1} \right| = \frac{1}{3} \ln 2.$$

$$-\frac{\ln 2}{3} - \frac{\ln 2}{3} = -\frac{2 \ln 2}{3}$$

Answer: $-\frac{2\ln 2}{3} \approx -0.462$. (The negative sign makes sense: on $[4, 5]$, $x-3 > 0$ but $x-6 < 0$, so the integrand is negative throughout this interval.)